

JEE MAIN REVISION MASTER: MAGNETIC EFFECT OF CURRENT SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 209 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Biot-Savart Law & Ampere's Circuital Law	109 PYQs	Magnetic Field
Very High	Lorentz Force, Cyclotron & Magnetic Force on Wires	101 PYQs	Magnetic Forces

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Biot-Savart Law & Ampere's Circuital Law	$dB = \frac{\mu_0 I dl \sin\theta}{4\pi r^2}, \quad B_{\text{wire}} = \frac{\mu_0 I}{2\pi d}, \quad B_{\text{center}} = \frac{\mu_0 N I}{2R}, \quad B_{\text{solenoid}} = \mu_0 n I$
Lorentz Force, Cyclotron & Magnetic Force on Wires	$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}), \quad r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}, \quad T = \frac{2\pi m}{qB}, \quad \frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Biot-Savart Law & Ampere's Circuital Law (Very High Yield — 109 PYQs)

Core Principle: Calculation of magnetic fields generated by steady currents in wires, loops, and solenoids.

■■ **Caution / JEE Pitfall:** In solenoid formula $B = \mu_0 n I$, n is number of turns per UNIT LENGTH ($n = N/L$), not total turns N .

■ **Speed Trick:** Right-hand thumb rule: thumb points along current, curled fingers give direction of circular magnetic field lines.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "An insulated wire is wound so that it forms a flat coil with $N=200$ turns. The radius of the innermost turn is $r_1=3\text{cm}$, and of the out..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "The magnetic field at the centre of a current carrying circular loop of radius R is $16\mu\text{ T}$. The magnetic field at a distance $x=\sqrt{\dots}$ " [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "An infinitely long straight wire carrying current I is bent in a planar shape as shown in the diagram. The radius of the circular part is ..." [Ans: B]

3.2 Lorentz Force, Cyclotron & Magnetic Force on Wires (Very High Yield — 101 PYQs)

Core Principle: Trajectory of charged particles in magnetic fields, velocity selector ($\vec{E} \perp \vec{v}$), and force between parallel currents.

■ **Caution / JEE Pitfall:** Magnetic force $\vec{F} = q(\vec{v} \times \vec{B})$ does ZERO work because $\vec{F} \perp \vec{v}$. It changes direction of velocity, never kinetic energy!

■ **Speed Trick:** Parallel currents in the SAME direction ATTRACT; anti-parallel currents REPEL.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "A current carrying circular loop of radius 2 cm with unit normal $\hat{n} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ is placed in a magnetic field, $\vec{B} = 4\hat{i} + 3\hat{j}$ T. The torque experienced by the loop is _____ N·m. [Ans: D]"
- **[2026 Evening Shift] Q#2 (MCQ):** "A particle having charge 10^{-9} C moving in x - y plane in fields of $0.4\hat{j}$ N/C and $4 \times 10^{-3}\hat{k}$ T experiences a force $\vec{F}$. [Ans: A]"
- **[2026 Morning Shift] Q#2 (MCQ):** "Three long straight wires carrying current are arranged mutually parallel as shown in the figure. The force experienced by 15 cm length of wire 2 is _____ N. [Ans: B]"

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A current carrying circular loop of radius 2 cm with unit normal $\hat{n} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ is placed in a magnetic field, $\vec{B} = B_0(3\hat{i} + 2\hat{j})$. If $B_0 = 4 \times 10^{-3}$ T and current $I = 100\sqrt{2}$ A, the torque experienced by the loop is _____ Wb·A. ($\pi = 3.14$)

- (A) 16×10^{-5} kN·m
- (B) 5024×10^{-7} kN·m
- (C) 5024×10^{-7} iN·m
- (D) 5024×10^{-7} jN·m [Correct]

Correct Answer: D

Solution Analysis: When a current-carrying loop is placed in an external magnetic field, it experiences a torque ($\vec{\tau}$). The magnetic properties of the loop are represented by its magnetic dipole moment ($\vec{m}$), defined as: $\vec{m} = I\vec{A}$. Where I is the current and $\vec{A}$ is the area vector of the loop. The direction of the area vector is perpendicular to the plane of the loop, given by the unit normal $\hat{n}$. $\vec{A} = A\hat{n}$. The torque is given by the cross product of the magnetic moment and the magnetic field: $\vec{\tau} = \vec{m} \times \vec{B} \Rightarrow |\vec{\tau}| = IAB \sin \theta$

Benchmark Problem 2: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

An insulated wire is wound so that it forms a flat coil with $N=200$ turns. The radius of the innermost turn is $r_1=3\text{cm}$, and of the outermost turn $r_2=6\text{cm}$. If 20 mA current flows in it then the magnetic moment will be $\alpha \times 10^{-2} \text{ A}\cdot\text{m}^2$. The value of α is ____.

- (A) 4.4
- (B) 2.64 **[Correct]**
- (C) 3.25
- (D) 1.2

Correct Answer: B

Solution Analysis: For a single circular turn of radius r carrying current I , the magnetic moment is : $m = I \cdot A = I(\pi r^2)$ In a flat coil where the turns are wound from r_1 to r_2 , the number of turns per unit radial length is $n = \frac{N}{r_2 - r_1}$. The total magnetic moment (M) is the integral of the moments of all infinitesimal turns of thickness dr : $M = \int_{r_1}^{r_2} (I\pi r^2) \cdot n dr \Rightarrow M = I\pi n \int_{r_1}^{r_2} r^2 dr = I\pi \left(\frac{N}{r_2 - r_1} \right) \left[\frac{r^3}{3} \right]_{r_1}^{r_2} \Rightarrow M = \frac{I\pi N}{3} (r_2^3 - r_1^3)$
Using t